Laboratory Exercise 05

Topics

1. Using conditional constructs to make decisions (Review of Lab03)
   a. Processing and comparisons between variables with `int`, `float` and `str` values
   b. Boolean expressions
   c. Logical choices
   d. Conditional constructs (`if`, `else`, and `elif`)
   e. Nested Blocks and Statements
2. Using loops for repeated execution of instructions (Review Lab04)
   a. Loops with `while` and `for` statements
   b. Processing complex inputs
   c. Simulations

Discussion

I. What is a good strategy for creating a solution to a complex problem?
II. How can you tell if the program that implements the solution is working correctly?

Exercises (in red = suggested ones to complete during the lab)

Part 1 – Summary exercises

***Goal**: For each of the following exercises, write a Python program that responds to the above requests. Complete at least two exercises during the lab session, and the remaining ones at home.*

05.1.1 ATM. When you use an Automatic Teller Machine (ATM) with your debit or credit card, you must use a Personal Identification Number (PIN) to access your account. If a user misses the PIN three times, the machine holds the card and locks it. Assuming that the user's PIN is `1234`, write a program that asks the user to type in the PIN, allowing a maximum of three attempts. The program should work like this:

I. when the user enters the correct number, they display the message `'Your PIN is correct'` and it terminates;
II. when the user enters a wrong number, the program displays the message `'Your PIN is incorrect'` and, if it asked for the PIN less than three times, it asks again;
III. When the user enters a wrong PIN three times, the program displays the message `'Your bank card is blocked'` and it terminates.

[P3.39]

05.1.2 Noms des pays. In French, the names of nations are feminine when they end with the letter *e*, otherwise they are masculine, with the exception of the following nouns, which are masculine even if they end with e:

I. *le Belize*;
II. *le Cambodge*;
III. *le Mexique*;
IV. *le Mozambique*;
V. *le Zaïre*;

VI. *le Zimbabwe*.
VII. *les Etats-Unis*;
VIII. *les Pays-Bas*.

Write a program that takes the name of a country in French and adds the article: `'le'` for masculine nouns and `'la'` for feminine nouns, such as `'le Canada'` or `'la Belgique'`. If, however, the name of the country begins with a vowel, the article becomes `'l'` (e.g., `'Afghanistan'`). Finally, for the countries listed here, which have a plural noun, the article `'les' is used`:

[P3.30]

[SUGGESTED] 05.1.3 Integer factorization. Write a program that asks the user for an integer and displays its prime factors. For example, if the user provides the number `150`, the program should display:

```
2
3
5
5
```

[P4.16]

05.1.4 At the cinema. Write a program that handles the presale of a limited number of movie tickets. Each buyer can buy a maximum of `4` tickets and no more than `100` can be sold in total. The program must ask the user how many tickets they want to buy, and then display the number of tickets left. The operation must be repeated until the tickets are sold out, displaying the number of buyers at the end. [P4.33]

Part 2 – Applications

Delivery: *For each of the following exercises, write a Python program that responds to the above requests. Complete at least one exercise during the exercise, and the rest at home.*

[SUGGESTED] 05.2.1 Random Number Generator. Using the formula $r_{new} = (a \times r_{old} + b) \% m$, where a, b, r, and m are integers, and then repeating the calculation by assigning the current value of r_{new} to r_{old} , we obtain a simple random number generator. Write a program that asks the user to provide an initial value for r_{old} (this value is the *seed* of the random number generator), and then displays the first `100` integers generated, using `a = 32310901`, `b = 1729`, and `m =` 2^{24}. [P4.27]

05.2.2 The Drunken Walk. A drunk person walks on a grid of streets. At each intersection, he randomly chooses one of the four directions, walks to the next intersection, and repeats the same random choice. You might think that, overall, the person won't move much, because overall the movements related to these random choices will cancel each other out. But is this really the case? Write a program that represents the position of the person in the grid of streets as pairs of integers (x, y). Implement the drunk walking algorithm described above by considering `100` intersections and `(0, 0)` as the starting position. [P4.24]

05.2.3 Prey and predator. The Lotka-Volterra equations describe a predator-prey ecological model that is based on a set of fixed positive constants:

A. the growth rate of prey;
B. the rate of destruction of prey by predators;
C. the mortality rate of predators;
D. the rate of increase in predators through the consumption of prey.

Given these constants, the following conditions apply in this model:

I. a population x of prey increases at a rate $dx = Ax\,dt$ (proportional to the number of prey) but is simultaneously destroyed by predators at a rate $dx = -B\,xy\,dt$ (proportional to the product of the numbers of prey and predators);
II. a population of predators y decreases at a rate $dy = -C\,y\,dt$ (proportional to the number of predators), but increases at a rate $dy = D\,xy\,dt$ (proportional to the product of the number of prey and predators).

From these conditions, the following system of differential equations is derived:

$$dx/dt = Ax - Bxy$$

$$dy/dt = -Cy + Dxy$$

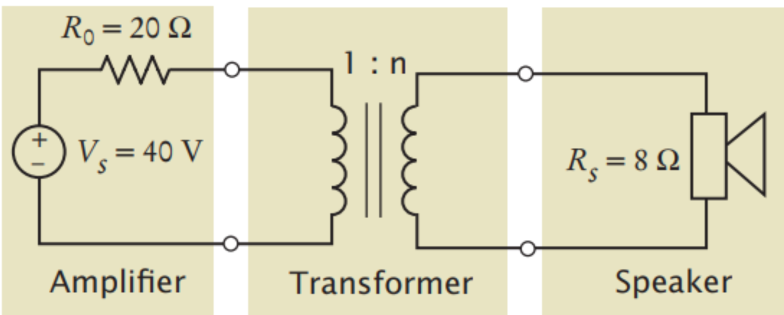
This means that, considering two time periods n and $n+1$, the change in populations sizes of, respectively, prey (x) and predators (y), from one period to the next is given by:

$$x_{n+1} = x_n \times (1 + A - B * y_n)$$

$$y_{n+1} = y_n \times (1 - C + D * x_n)$$

Write a program that asks the user to input the value of the four constants, the initial numbers of the prey and predator populations, and the number of periods you want to simulate, and then calculates and visualizes the two populations sizes in each of the periods considered. As test inputs, use `A = 0.1`, `B = 0.01`, `C = 0.01`, and `D = 0.00002`, with initial populations of prey and predators numbering `1000` and `20`, respectively. [P4.36]

[SUGGESTED] 05.2.4 Transformer. Transformers (the electrical components, not the movie robots 😉) are often constructed by wrapping coils of wire around a ferrite core. The figure illustrates a situation that occurs in various audio devices such as mobile phones and music. In this circuit, a transformer is used to connect a speaker to the output of an audio amplifier.



The symbol used to represent the transformer is intended to suggest two coils of wire. The transformer parameter **n** is called the transformer's "turns ratio." (The number of times a wire is wound around the core to form a coil is called the "number of turns" of the coil. The turns ratio is literally the ratio of the number of turns in the two coils of wire). When designing the circuit, engineers are mainly concerned with the value of the power delivered to the speakers: this power ensures that the speakers produce the sounds we want to hear with the desired loudness. Suppose you connect the speakers directly to the amplifier without using the transformer. Some of the power available from the amplifier would go to the speakers. The rest of the available power would be lost in the amplifier itself. The transformer is added to the circuit to increase the fraction of amplifier power that is delivered to the speakers. The power delivered to the speakers is calculated with the formula:

$$P_s = R_s \left(\frac{nV_s}{n^2R_0 + R_s}\right)^2$$

Write a program that models the circuit shown and varies the turns ratio from `0.01` to `2` in `0.01` increments, then determine the value of the turns ratio that maximizes the power output to the speakers. [P4.40]